A063 v0.2.1 — Hicks invariant-surface qualification

Analytical reference (selected T34 target); parent v0.2.0 preserved; no v1.0.7 certification.

New gates: 7 PASS / 2 FAIL / 3 UNRESOLVED.

Scientific question and preregistered hypotheses

Does the closed $T(3,4)$ Beltrami vortex-line orbit admit a finite smooth stationary material/vorticity surface whose *solid-torus core* is also knotted, and can this be certified as nonlinearly stable under coupled perturbed 3D Euler evolution? We contrast two constructions: (i) the previously failed constant-radius tube centered on the knot and (ii) regular stationary Hicks level sets $\psi = \psi_0$. The latter are not assumed to be knotted tubes.

$$\psi(s,z) = A\big[J(\lambda r/a) - (r/a)^2 J(\lambda)\big]s^2/r^2, \tag{1}$$

$$J(x) = \frac{\sin x}{x} - \cos x, \quad r = \sqrt{s^2+z^2}, \quad \lambda = 4.4934094579, \tag{2}$$

$$u_s = -\frac{1}{s}\partial_z\psi, \quad u_z = \frac{1}{s}\partial_s\psi, \quad u_\phi = \frac{\lambda\psi}{as}, \quad \boldsymbol{\omega} = (\lambda/a)\mathbf{u}. \tag{3}$$

Therefore $\mathbf{u}\cdot\nabla\psi = 0$ and $\boldsymbol{\omega}\cdot\nabla\psi = 0$ for the exact first-Beltrami Hicks branch, wherever $\nabla\psi \neq 0$. We independently evaluate numerical normals using centered derivatives of analytic ψ against the separately evaluated analytic velocity.

1. Explicit regular level surfaces

The vortex-line initial position $s_0/a = 0.9683098302$, and its reconstructed level is $\psi_0 = 0.138575114976$. We solve $\psi(s_c + r(\theta)\cos\theta, r(\theta)\sin\theta) = q\psi_0$ for $q \in \{0.9, 1, 1.1\}$, with 96, 192 and 384 meridional samples. The radial solutions are single-valued and strictly positive in the tested intervals. Revolving each curve around the symmetry axis yields a regular embedded torus.

ψ/ψ_0	$s_{\min}/a$	sphere clearance $/a$	W	max $\lvert u_n\rvert/u_{\rm rms}$
0.90	0.13882	0.02850	0.747563717	5.909e-08
1.00	0.14666	0.03169	0.750000000	5.900e-08
1.10	0.15418	0.03488	0.752329435	5.887e-08

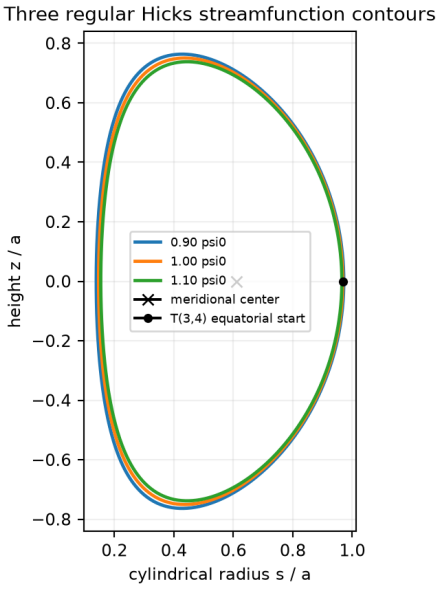


Figure 1: Closed nested meridional ψ contours. Each avoids the axis $s = 0$ and the sphere boundary.

2. Topology: unknotted invariant torus, knotted line ON the torus

The meridional contours are star-shaped about $s = s_c, z = 0$, and can be continuously deformed to a small circle by $r_t(\theta) = (1-t)r(\theta) + tr_*$ without touching $s = 0$. Revolution constructs an explicit isotopy to the **standard, unknotted solid torus**. Its core is a planar circle $C(\phi) = (s_c \cos\phi, s_c \sin\phi, 0)$, an unknot. In contrast, the original $T(3,4)$ trajectory winds three times azimuthally and four times meridionally on the *boundary* $\psi = \psi_0$. The torus is not a knotted tubular neighborhood whose solid-torus core is $T(3,4)$.

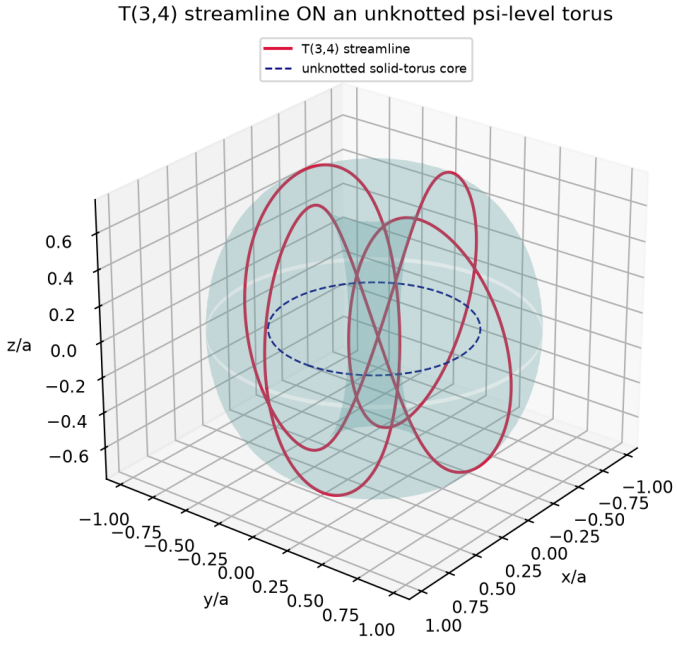


Figure 2: An invariant standard torus (translucent), the red knotted vortex line, and the dashed unknotted circle that is the solid-torus core.

3. Material and vorticity flux

The exact invariant surface has zero normal flux. A separate numerical check yields the normal-velocity values above; Richardson-like reduction across finite-difference steps checks convergence to zero. Since $\omega = (\lambda/a)u$, the exact vorticity-normal flux is zero too. This is **not** a localized vortex tube: the volume enclosed is a standard solid torus and the original Hicks flow has distributed vorticity throughout the spherical vortex aggregate.

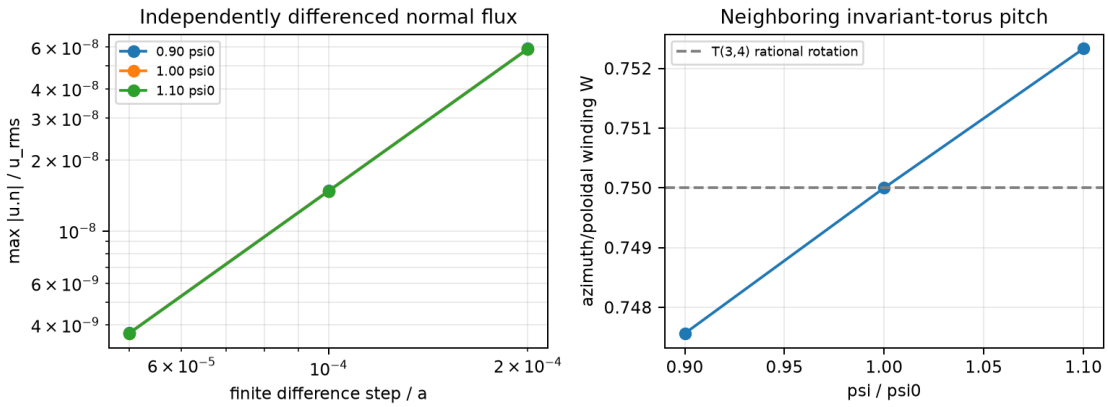


Figure 3: FD-normal flux convergence (left) and invariant-torus winding shear (right). Winding shear is kinematic and is not a 3D Euler instability growth rate.

4. Frozen parent evidence and continuation semantics

Parent v0.2.0 gate statuses were loaded from byte-hashed files, checked against its original gate ledger and source-seal hash, and *not recomputed or rewritten*. The failed steady constant-normal-radius tube (T5 FAIL) and constant rotation of neighbor lines (T6 FAIL) remain historical results. A v0.2.1 S6 FAIL specifically rejects *the psi-level torus as a knotted-core construction*; it is not a proof that no other knotted material surface could exist.

5. New gate ledger

Gate	Status	Qualification
S0	PASS	Are frozen parent v0.2.0 hashes and FAIL/UNRESOLVED status immutable?
S1	PASS	Does fresh integration reproduce frozen T(3,4) reference line?
S2	PASS	Do explicit closed regular psi-level contours occur at three frozen levels?
S3	PASS	Does the original T(3,4) line sit on the reconstructed psi-level torus?
S4	PASS	Are u and omega tangent to every regular psi-level surface?
S5	PASS	Are these invariant level-set boundaries standard unknotted tori of revolution?
S6	FAIL	Does the psi-level solid torus itself have a T(3,4)-knotted CORE?
S7	FAIL	Does a finite interval of invariant psi levels share identical 3:4 line closure?
S8	PASS	Are invariant level contours numerically converged and regular?
S9	UNRESOLVED	Does v0.2.1 demonstrate an isolated finite T(3,4) vortex core stable in 3D Euler?
S10	UNRESOLVED	Has native C++ FP64 v1.0.7 certification independently occurred?
S11	UNRESOLVED	Can local shear/deformation establish dynamical instability without perturbed Euler?

6. Remaining scientific blockers

A complete falsifier of general finite-core vortex-tube persistence must construct a localized finite-vorticity field, a nonaxisymmetric material boundary with independently checked knot type, and a *coupled 3D incompressible Euler perturbation/pressure evolution*. Register perturbations and grid resolutions in advance; monitor div(u), energy, helicity, knot/link invariants, tube flux, boundary clearance and error convergence for at least ten core periods. None of these certifications was executed in v0.2.1. This is an analytical/numerical `REFERENCE_ONLY` package, not an official v1.0.7 FREEZE/CERTIFY instance. **Conclusion.** Smooth stationary Hicks ψ-tori *do* exist and are material/vorticity-tangent. They are unknotted ordinary tori of revolution. A $T(3,4)$ trajectory on such a surface does not supply a stable isolated $T(3,4)$ vortex-tube core. General finite-core existence and time-dependent stability remain **UNRESOLVED**.

References

W. M. Hicks, *Researches in Vortex Motion III* (1899), doi:10.1098/rsta.1899.0002. A063 v0.2.0 local archival evidence, byte-bound to v0.2.1 in `baseline/`.